



RUPA RAJA'S

PCMR E.M. SCHOOL

The Next Generation School

Bathalapalli Road, Nagaluru - 515678, Dharmavaram, Sri Sathya Sai (Dist) Ph : 9989577577

CLASS - 10

CLICK
ME

Name of the Student: _____

Sec : Challengers

Sub: MATHS

Time: 3 hours

Marks : 100

**GRAND
TEST 2****SECTION - I****Note : i. Answer all the questions in one word (or) Phrase.****12 x 1 = 12m****ii. Each question carries 1 mark.**

1. If H.C.F (a,20) = 2, L.C.M (a,20) = 60, then a = _____

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A. 2**B. 6****C. 12****D. 24**2. The zero of the polynomial of $p(x) = 4x - 8$ is

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A. 2**B. -6****C. -5****D. 5**

3. Write a polynomial of degree 2, having constant term 2026.

4. Create a linear equation which is parallel to the line $3x - 2y + 4 = 0$.

5. Match the following.

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q) Sum of first 10 natural numbers.**i. 110****r) Sum of first 10 odd natural numbers****ii. 55****s) Sum of first 10 even natural numbers****iii. 100****A. p-i, q-ii, r-iii****C. p-iii, q-i, r-ii****B. p-ii, q-iii, r-i****D. p-i, q-iii, r-ii**6. If $\operatorname{Cosec}\theta - \cot\theta = x$ then $\operatorname{Cosec}\theta + \cot\theta =$ _____

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a) $-x$ b) $x+1$ c) $-\frac{1}{x}$ d) $+\frac{1}{x}$ 7. **Statement 1** : All similar triangles are congruent**Statement 2** : All Right angled isosceles triangle are similar.

a) Both statements are true.

b) Statement 1 is true, statement 2 is false.

c) Statement 1 is false, statement 2 is true.

d) Both statements are false.

8. Define line of sight.

9. If the volume of a cube is 125cm^3 , then its side is _____**a) 6 cm****b) 12 cm****c) 10 cm****d) 5 cm**

10. $P(E) = 0.05$, what is the probability of 'not E' _____
11. Draw a circle and two lines parallel to a given line such that one is a tangent and the other a secant to a circle.
12. Form a quadratic equation whose roots are 2 and 3.

SECTION - II

Note : i. Answer all the questions

8 x 2 = 16m

ii. Each question carries 2 marks.

13. Create a quadratic polynomial whose sum and product of zeros are $\frac{1}{4}$ and -1 respectively.
14. Find the nature of roots of $2x^2 + 3x + 5 = 0$.
15. Give an example for
 - i. Similar figures and
 - ii. Non – similar figures.
16. If $\sin(A - B) = \frac{1}{2}$, $\cos(A + B) = \frac{1}{2}$, $0^\circ < A + B \leq 90^\circ$, $A > B$, find A and B.
17. Define angle of depression with simple rough diagram.
18. From a point Q, the length of the tangent to a circle is 24cm and the distance of from the centre is 25cm. Find the radius of the circle.
19. Consider the following situation in each find out whether you need volume or area and why?
 - i. Quantity of water inside a bottle.
 - ii. Canvas needed for making a tent.
20. Find the coordinates of the point which divides the line joining (-1, 7) and (4, -3) in the ratio 2:3.

SECTION - III

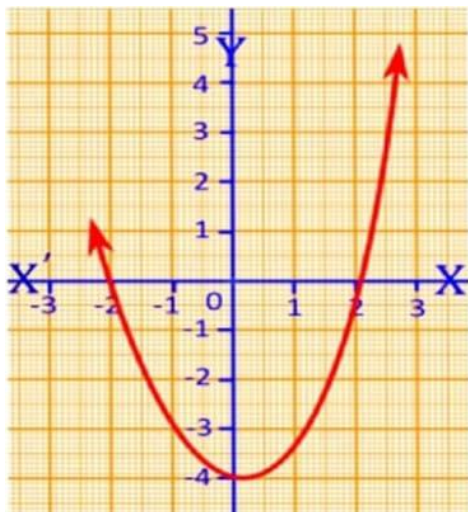
Note : i. Answer all the questions

8 x 4 = 32m

ii. Each question carries 4 marks.

21. One card is drawn from a well –shuffled deck of 52 cards. Create 4 questions based on this?
“ Find the probability that the drawn card will be the king. ”
22. Write the formula to find mode of a grouped data and explain the terms in it.
23. 2 cubes each of volume 64cm^3 are joined end to end. Find the surface area of the resulting cuboid?

24. Find the roots of quadratic equation $3x^2 - 4\sqrt{3}x + 4 = 0$.
25. Write $\cos A$, $\tan A$ and $\sec A$ in terms of $\sin A$.
26. If sum of first 14 terms of an A.P is 1050 and its first term is 10, then find 20th term.
27. Prove that the length of tangent drawn from an external point to a circle are equal.
28. Answer the following questions by observing the graph.



- a. Name the shape of the graph?
- b. Write the number of zeros of the polynomial.
- c. Write the zeroes of the polynomial.
- d. Write the product of zeros of the polynomial.

SECTION - IV

Note : i. Answer all the questions

5 x 8 = 40m

ii. Each question carries 8 marks.

iii. There is an internal choice for each question.

29. **A.** is $\sqrt{5}$ irrational? Justify your answer.

(Or)

B. State and prove basic proportionality theorem.

30. **A.** Two students claim to have found the points of trisection of the line segment joining A (2, -2) and B (-7,4) as follows.

Bharath : (-1,0) and (-4,2)

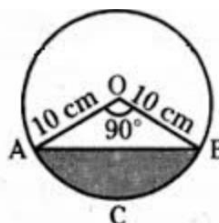
Vinod : (1,0) and (4,-2). Who is correct? Justify?

(Or)

B. A Chord of a circle of radius 10cm subtends a right angle at the centre. Find the area of the corresponding.

i. minor segment

ii. major sector (use $\pi = 3/14$)



31. **A.** one card is drawn from a well – shuffled deck of 52 cards. Find the probability of getting.

i. **A king of red colour**

ii. **A face card**

iii. **A black face card**

iv. **The jock of diamond**

v. **A spade**

vi. **The queen of diamond.**

(Or)

B. The angle of elevation of the top of a tower from a point on the ground is 30° on moving 20m nearer to the tower, the angle of elevation increases to 45° . Find the height of the tower and the distance of the first point from the tower.

32. **A.** The following frequency distribution gives the monthly consumption of electricity of 68 consumers of a locality. Find the median.

Monthly consumption	65-85	85-105	105-125	125-146	145-165	165-185	185-205
Number of consumers	4	5	13	20	14	8	4

(Or)

B. 200 logs are stacked in the manner that is 20logs in the bottom row, 19 in the next row, 18 in the row next to it and 50 on in how many rows are the 200 logs placed and how many logs are in the top row?

33. **A.** draw the graph of the following pair of linear equations $2x + 3y - 6 = 0$ and $x + y + 3 = 0$.

(Or)

B. 10 students of class x took part in a mathematics quiz if the number of girls is 4 more than the number of boys, find the number of boys and girls who took part in the quiz by graphically.